

Target Validation Using Multi-Omics Evidence

Objective

The objective of this task was to validate the top-ranked therapeutic target identified during the prioritization phase using independent **multi-omics datasets**, thereby reducing biological and translational risk prior to downstream drug discovery efforts. Validation was performed using **transcriptomic and functional genomics evidence** derived from large-scale public cancer resources.

The target selected for validation was **Epidermal Growth Factor Receptor (EGFR)** in **Lung Adenocarcinoma (LUAD)**.

Rationale for Target Selection

EGFR emerged as a top-priority candidate based on strong genetic association, oncogenic driver status, and established druggability. However, industry-level target validation requires confirmation that:

1. The target is **dysregulated in patient tumors**, and
2. Tumor cells demonstrate **functional dependency** on the target for survival.

Accordingly, EGFR was evaluated using complementary **transcriptomic (TCGA)** and **functional genomics (DepMap CRISPR)** datasets.

Datasets and Validation Strategy

Transcriptomic Validation (TCGA-LUAD)

- **Data source:** TCGA via UCSC Xena
- **Data type:** RNA-seq gene expression (HiSeqV2)
- **Samples analyzed:**
 - Tumor: 515
 - Normal: 59

EGFR expression levels were extracted and compared between tumor and normal tissues using non-parametric statistical testing (Mann–Whitney U test).

Functional Genomics Validation (DepMap)

- **Data source:** Broad Institute DepMap

- **Data type:** Genome-wide CRISPR knockout gene-effect scores
- **Cell lines analyzed:** 53 LUAD cell lines

Gene-effect scores quantify cellular dependency, where more negative values indicate stronger dependence on the target gene.

Dependency categories were defined as:

- **Strong dependency:** ≤ -0.5
 - **Moderate dependency:** -0.3 to -0.5
 - **Weak / none:** > -0.3
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Results

1. Transcriptomic Evidence

EGFR expression was detectable in both tumor and normal lung samples. Median expression values were comparable between the two groups, and statistical testing did not demonstrate a significant difference in expression levels.

Interpretation:

EGFR does not function primarily as an **overexpression-driven oncogene in LUAD**. Instead, its oncogenic role is driven by **activating mutations**, consistent with known clinical biology.

2. Functional Genomics Evidence

CRISPR dependency analysis revealed:

- **Strong dependency:** 9 cell lines (17%)
- **Moderate dependency:** 7 cell lines
- **Weak / no dependency:** 37 cell lines

The median EGFR gene-effect score was -0.187 .

Interpretation:

A biologically meaningful subset of LUAD cell lines exhibits strong dependence on EGFR, indicating **oncogene addiction in specific molecular contexts** rather than uniform dependency across all tumors.

Integrated Biological Interpretation

The combined multi-omics evidence supports the following conclusions:

- EGFR activity in LUAD is **mutation-driven rather than expression-driven**.
- Functional dependency is **context-specific**, aligning with known EGFR-mutant tumor biology.
- The heterogeneity observed across cell lines reflects real-world clinical variability in therapeutic response.

This concordance between transcriptomic neutrality and functional dependency strongly supports EGFR's established role as a **precision oncology target**.

Validation Conclusion (Go / No-Go)

Decision: GO (Context-Specific)

EGFR is validated as a therapeutically actionable target in LUAD under defined molecular contexts, particularly in tumors harboring activating EGFR mutations.

While not universally essential across all LUAD tumors, the presence of strong functional dependency in a clinically relevant subset supports continued drug discovery and optimization efforts, especially for:

- resistance mechanisms
 - next-generation inhibitors
 - combination therapy strategies
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Key Limitations

- Transcriptomic data does not capture mutation-specific activation.
 - CRISPR dependency reflects in vitro cellular models and may not fully represent tumor microenvironment effects.
 - Patient stratification based on mutation status was not incorporated in this analysis.
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Overall Conclusion

This multi-omics validation confirms EGFR as a biologically and clinically relevant target in LUAD, characterized by mutation-dependent oncogenic activation and selective

cellular dependency. The findings support EGFR as a high-confidence, precision-guided oncology target suitable for downstream drug discovery and therapeutic optimization.

Feature Contribution Analysis (Top Prioritized Targets)

The feature contribution heatmap illustrates the multidimensional evidence profile supporting the top-ranked LUAD targets. Each row represents a gene, while each column corresponds to a specific biological or translational evidence source. Values are normalized between 0 and 1, allowing direct comparison across heterogeneous datasets.

EGFR — High-Confidence, Clinically Validated Target

EGFR demonstrates consistently high scores across nearly all evidence categories. Strong global association (0.83), cancer gene census inclusion (0.88), and IntOGen driver significance (0.90) confirm its causal role in LUAD tumorigenesis. The presence of substantial somatic mutation evidence (0.85) and biomarker relevance (0.83) further supports its clinical translatability. Importantly, measurable ChEMBL activity (0.64) confirms pharmacological tractability, distinguishing EGFR as a low-risk, fully validated oncology target with proven druggability.

KRAS — Genetically Strong but Historically Challenging Target

KRAS exhibits one of the strongest genetic driver profiles in the dataset, with high global association (0.80), IntOGen significance (0.91), and biomarker relevance (0.93). However, the absence of ChEMBL evidence reflects its historical classification as “undruggable,” despite recent advances. This pattern highlights KRAS as a biologically essential but technically challenging target, representing high biological value coupled with elevated development risk.

BRAF — Targetable Oncogenic Kinase with Moderate Evidence Breadth

BRAF shows robust oncogenic support through IntOGen (0.85), somatic mutation evidence (0.79), and biomarker relevance (0.95). However, comparatively lower cancer gene census and literature coverage indicate a more subtype-specific role in LUAD. This profile positions BRAF as a context-dependent target with strong therapeutic rationale in selected patient populations rather than broad applicability.

STK11 (LKB1) — Tumor Suppressor with Limited Direct Druggability

STK11 displays strong genetic and biomarker associations but minimal somatic activation evidence (0.19) and complete absence of ChEMBL druggability. This reflects its function as a tumor suppressor rather than an oncogenic driver. While biologically critical, STK11 is poorly suited for direct pharmacological modulation, highlighting an important distinction between biological importance and therapeutic feasibility.

TP53 — Central Cancer Gene with High Translational Constraints

TP53 demonstrates extensive genetic and literature support, consistent with its role as the most frequently mutated gene in human cancers. However, the absence of ChEMBL and Reactome tractability reflects the long-standing difficulty of directly targeting TP53. This reinforces TP53 as a mechanistically central but practically high-risk target, more suitable for indirect pathway-based strategies.

Cross-Target Interpretation

This heatmap reveals three critical insights:

1. **Biological relevance alone is insufficient** — several highly mutated genes lack therapeutic tractability.
 2. **EGFR uniquely combines genetic causality, clinical relevance, and druggability**, making it the most balanced and development-ready target.
 3. **High literature or mutation frequency does not equate to drug discovery feasibility**, emphasizing the importance of integrated scoring rather than single-evidence interpretation.
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Strategic Interpretation

The feature-level analysis demonstrates that the prioritization framework successfully discriminates between:

- **Low-risk, development-ready targets** (EGFR)
- **High-value but technically difficult targets** (KRAS, TP53)
- **Subtype- or context-dependent opportunities** (BRAF, STK11)

This multidimensional profiling ensures that final target selection is driven not only by biological importance, but by realistic translational and drug development considerations.

Final Conclusion

Among the top-ranked LUAD targets, **EGFR displays the most complete and balanced evidence fingerprint**, integrating genetic causality, oncogenic relevance, clinical validation, and pharmacological tractability. This reinforces its selection as the lead target for downstream functional validation and therapeutic development